Siddharth K

1. We know that $999 \equiv -1 \pmod{1000}$, and since $999 \equiv 9999 \equiv \cdots \equiv 99\ldots9 \pmod{1000}$, we have $9 \times 99 \times 999 \times \cdots \times 9\ldots9 \equiv 9 \times 99 \times (-1)^{997} \equiv 109 \pmod{1000}$. □

2. We can see that $1^2 \equiv 1 \pmod 8, 2^2 \equiv 4 \pmod 8, 3^2 \equiv 1 \pmod 8, 4^2 \equiv 0 \pmod 8, 5^2 \equiv 1 \pmod 8, 6^2 \equiv 4 \pmod 8, 7^2 \equiv 1 \pmod 8$, and $8^2 \equiv 0 \pmod 8$. From this, it follows that the possible remainders when squares are divided by 8 are $0, 1$, and 4. Since no combination of three numbers chosen from $0, 1$, and 4 can add up to 7 or any other number congruent to 7 modulo 8, n cannot be written as a sum of three squares. □

3. Since $13 \equiv -1 \pmod 7$, $3^n + 13 \cdot 10^n \equiv 3^n - 10^n \pmod 7$. We can see that $3^n - 10^n = (3-10)(3^{n-1}10^0 + \cdots + 3^0 10^{n-1}) = -7(3^{n-1}10^0 + \cdots + 3^0 10^{n-1})$ which is a multiple of 7. Therefore, $3^n + 13 \cdot 10^n$ is also a multiple of 7. □

4. Testing possible solutions for $x \equiv 5 \pmod 7$ and $x \equiv 7 \pmod{11}$, we can see that $x \equiv 40 \pmod{77}$. We can find that by adding multiples of 11 to 7 until we get a number that satisfies $x \equiv 5 \pmod 7$. Now, by the same process, we can test solutions for that and $x \equiv 3 \pmod 5$, and we get $x \equiv 348 \pmod{385}$. Finally, we can do the same for $x \equiv 348 \pmod{385}$ and $x \equiv 2 \pmod 3$ and we get $x \equiv 1118 \pmod{1155}$. □

5. We know that addition is well defined in $\mathbb{Z}/m\mathbb{Z}$, which means if $a \equiv b \pmod m$ and $c \equiv d \pmod m$, then $a + c \equiv b + d \pmod m$. If instead of $c \equiv d \pmod m$, we know that $-c \equiv -d \pmod m$, because $-1 \equiv -1 \pmod m$. That means $a + (-c) \equiv b + (-d) \pmod m$, or $a - c \equiv b - d \pmod m$. □

6. Here are the proofs: Divisibility by 9: We can rewrite the number as $n = 10^{n-1}d_n + 10^{n-2}d_{n-1} + \cdots + 10^0 d_1$, which we can assume is a multiple of 9. That means $10^{n-1}d_n + 10^{n-2}d_{n-1} + \cdots + 10^0 d_1 \equiv 0 \pmod 9$. We know that $10^n \equiv 1 \pmod 9$, so using the fact that we can add or multiply elements in $\mathbb{Z}/m\mathbb{Z}$, we get $n \equiv d_n + d_{n-1} + \cdots + d_1 \equiv 0 \pmod 9$. That shows us that the sum of the digits is also a multiple of 9. Therefore, if n is a multiple of 9, then so is the sum of its digits. We can go through those steps in reverse to show that any number that satisfies that condition is a multiple of 9, which proves that a number is a multiple of 9 if and only if the sum of its digits is a multiple of 9. □

Divisibility by 11: As before, we can rewrite the number as $n = 10^{n-1}d_n + 10^{n-2}d_{n-1} + \cdots + 10^0 d_1$. Assuming this number is divisible by 11, we get $n \equiv 0 \pmod{11}$. Also, note that $10^n \equiv (-1)^n \pmod{11}$, which alternates depending on whether n is even. This means $10^{n-1}d_n + 10^{n-2}d_{n-1} + \cdots + 10^0 d_1 \equiv d_1 - d_2 + d_3 \cdots (-1)^{n-1}d_n \equiv 0 \pmod{11}$, so if the alternating sum of the digits is a multiple of 11, n is as well. Reversing these steps would prove the converse, which means a number is a multiple of 11 if and only if the alternating sum of the digits is a multiple of 11. □

Divisibility by 7: Every number can be written as $n = 10a + b$ where a is some nonnegative integer and b is a one-digit nonnegative integer (which is also the units digit). Now let's assume $n \equiv 0 \pmod 7$. We know that $21b \equiv 0 \pmod 7$, so $10a + b - 21b \equiv 10a - 20b \equiv 10(a - 2b) \equiv 0 \pmod 7$. And $10(a - 2b)$ is only a multiple of 7 if $a - 2b$ is a multiple of 7. We can reverse those steps and combine them both to arrive at the conclusion that a number is a multiple of 7 if and only if the difference between the digits to the left of the units digit and twice the units digit is a multiple of 7. □

Divisibility by 13: Let's say $n = 10a + b$. Then $10a + b \equiv 10a + b + 39b \equiv 10a + 40b \equiv 10(a + 4b) \equiv a + 4b \pmod{13}$. Therefore, if $10a + b$ is a multiple of 13, $a + 4b$ is too. Reversing those steps yields the fact that $a + 4b \equiv 0 \pmod{13} \implies 10a + b \equiv 0 \pmod{13}$. So adding four times the units digit to the rest of the number and seeing if the result is a multiple of 13 will tell you whether or not the original number is a multiple of 13. □

7. We are given that $f(r + i) \equiv 0 \pmod m$ for $0 \leq i \leq m - 1$. Since $f(x)$ is a polynomial function, it only consists of additions and multiplications of x and other constants. This means, for each remainder of x modulo m, the remainder when $f(x)$ is divided by m is 0. Since we know that the integers from r to $r + m - 1$ cover all possible remainders modulo m, $f(x) \equiv 0 \pmod m$ for all integers x. □

8. To prove this, I tried to find a way to construct an $n \times n$ square of blocked points and then show that using that method to construct such a square would make all points inside it blocked. So I started off by trying to find an example of a 3×3 square that satisfies the constraints. However, I couldn't find a way to do that.